

練習題 14.3

3. 證:

$$(1). \text{L.H.S} = J(c\vec{f}) = \cancel{\left(\frac{\partial f_i(\vec{x})}{\partial x_j} \right)} = \left(\frac{\partial (cf_i(\vec{x}))}{\partial x_j} \right)_{m \times n}$$

$$= \left(c \cdot \frac{\partial f_i(\vec{x})}{\partial x_j} \right)_{m \times n} = c \left(\frac{\partial f_i(\vec{x})}{\partial x_j} \right)_{m \times n} = cJ(\vec{f})$$

$$(2) \text{L.H.S} = J(\vec{f} + \vec{g}) = \left(\frac{\partial (f_i(\vec{x}) + g_i(\vec{x}))}{\partial x_j} \right)_{m \times n}$$

$$= \left(\frac{\partial f_i(\vec{x})}{\partial x_j} + \frac{\partial g_i(\vec{x})}{\partial x_j} \right)_{m \times n} = \left(\frac{\partial f_i(\vec{x})}{\partial x_j} \right)_{m \times n} + \left(\frac{\partial g_i(\vec{x})}{\partial x_j} \right)_{m \times n}$$

$$= J\vec{f} + J\vec{g}$$

$$(3) \text{當 } m=1 \text{ 時, } J(fg) = \left(\frac{\partial (fg)}{\partial x_1}, \frac{\partial (fg)}{\partial x_2}, \dots, \frac{\partial (fg)}{\partial x_n} \right)$$

$$= \left(\frac{\partial f}{\partial x_i} g + \frac{\partial g}{\partial x_i} f \right)_{1 \times n} = g \left(\frac{\partial f}{\partial x_i} \right)_{1 \times n} + f \left(\frac{\partial g}{\partial x_i} \right)_{1 \times n}$$

$$= gJf + fJg$$

(4) 當 $m > 1$ 時,

$$\text{L.H.S} = J(\vec{f}, \vec{g}) = J\left(\sum_{k=1}^m f_k g_k\right) = \sum_{k=1}^m J(f_k g_k)$$

$$= \sum_{k=1}^m (g_k Jf_k + f_k Jg_k) = \sum_{k=1}^m (g_k Jf_k) + \sum_{k=1}^m (f_k Jg_k)$$

$$= \vec{g}J\vec{f} + \vec{f}J\vec{g} = \text{R.H.S}$$

$$4. (J\vec{f}, \vec{f}) = \sum_{k=1}^n \left(\frac{df_k(x)}{dx} \cdot f_k(x) \right) = \frac{1}{2} \sum_{k=1}^n \frac{d f_k^2(x)}{dx} = \frac{1}{2} \cdot \frac{d}{dx} \left(\sum_{k=1}^n f_k^2(x) \right) = \frac{1}{2} \frac{d}{dx} \|\vec{f}(x)\|^2 = 0$$

幾何解釋: 在半徑為 $\|\vec{f}(t)\| = R > 0 (t \in [a, b])$ 的球面 ~~上~~

已知 $\vec{f} \in \pi$, 任取一點 $\vec{f}(t)$, $J\vec{f}(t)$ 便是過 $\vec{f}(t)$ 點的其中一條切線的方向向量 (相對球面)

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5. 由題意知.

$$\vec{f}(x, y, z) = \begin{pmatrix} f_1(x) \\ f_2(y) \\ f_3(z) \end{pmatrix}, \text{ 又知 } \frac{df_1(x)}{dx} = \alpha(x), \frac{df_2(y)}{dy} = \beta(y), \frac{df_3(z)}{dz} = \gamma(z)$$

$$\text{取 } f_1(x) = \int_0^x \alpha(t) dt, \quad f_2(y) = \int_0^y \beta(t) dt, \quad f_3(z) = \int_0^z \gamma(t) dt$$

$$\text{即 } \vec{f}(x, y, z) = \begin{pmatrix} \int_0^x \alpha(t) dt \\ \int_0^y \beta(t) dt \\ \int_0^z \gamma(t) dt \end{pmatrix}$$

$$6. (1) \text{ 取 } \lambda = \mu = 0, \quad \vec{f}(\vec{0}) = \vec{f}(0 \cdot \vec{x} + 0 \cdot \vec{y}) = 0 \cdot \vec{f}(\vec{x}) + 0 \cdot \vec{f}(\vec{y}) = \vec{0}$$

$$(2) \text{ 取 } \lambda = -1, \mu = 0, \quad \vec{f}(-\vec{x}) = -1 \cdot \vec{f}(\vec{x}) = -\vec{f}(\vec{x})$$

$$(3) \forall \vec{x} \in \mathbb{R}^n, \quad \vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = I \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \sum_{k=1}^n x_k \vec{e}_k$$

$$\vec{f}(\vec{x}) = \vec{f}\left(\sum_{k=1}^n x_k \vec{e}_k\right) = \sum_{k=1}^n x_k \vec{f}(\vec{e}_k) = x_1 \vec{f}(\vec{e}_1) + x_2 \vec{f}(\vec{e}_2) + \dots + x_n \vec{f}(\vec{e}_n)$$

7.

$$J\vec{f} = \left(\frac{\partial f_i(\vec{x})}{\partial x_j} \right)_{m \times n} \quad \vec{f}(\vec{x} + \vec{h}) - \vec{f}(\vec{x}) = \vec{f}(\vec{h}) = \begin{pmatrix} f_1(\vec{e}_1) & f_1(\vec{e}_2) & \dots & f_1(\vec{e}_n) \end{pmatrix} \vec{h}$$

$$\text{因此, } J\vec{f} = \begin{pmatrix} f_1(\vec{e}_1) & f_1(\vec{e}_2) & \dots & f_1(\vec{e}_n) \end{pmatrix}$$

$$8. \forall \vec{x}, \vec{y} \in \mathbb{R}^n \text{ 且 } \lambda, \mu \in \mathbb{R}, \quad \vec{E}(\lambda \vec{x} + \mu \vec{y}) = \lambda \vec{x} + \mu \vec{y} = \lambda \vec{E}(\vec{x}) + \mu \vec{E}(\vec{y})$$

即 $\vec{E}$ 是 $\mathbb{R}^n$ 中的恒等线性映射

$$J\vec{E} = \begin{pmatrix} \vec{E}(\vec{e}_1) & \vec{E}(\vec{e}_2) & \dots & \vec{E}(\vec{e}_n) \end{pmatrix} = \begin{pmatrix} \vec{e}_1 & \vec{e}_2 & \dots & \vec{e}_n \end{pmatrix} = I_n$$

練習題 14.4

$$1. \left(\frac{\partial u}{\partial x} \quad \frac{\partial u}{\partial y} \right) = \left(\frac{\partial u}{\partial(x^2+y^2)} \quad \frac{\partial u}{\partial(x^2+y^2)} \right) \begin{pmatrix} 2x & 2y \\ 2x & 2y \end{pmatrix} = \left(4x \frac{\partial u}{\partial(x^2+y^2)} \quad 4y \frac{\partial u}{\partial(x^2+y^2)} \right)$$

$$\Rightarrow \frac{1}{4x} \cdot \frac{\partial u}{\partial x} = \frac{\partial u}{\partial(x^2+y^2)} = \frac{1}{4y} \frac{\partial u}{\partial y}$$

$$\Rightarrow y \frac{\partial u}{\partial x} - x \frac{\partial u}{\partial y} = 0$$

$$2. \left(\frac{\partial u}{\partial x} \quad \frac{\partial u}{\partial y} \right) = \left(\frac{\partial u}{\partial(xy)} \quad \frac{\partial u}{\partial(xy)} \right) \begin{pmatrix} y & x \end{pmatrix}$$

練習題 14.4

$$1. \left(\frac{\partial u}{\partial x} \quad \frac{\partial u}{\partial y} \right) = \frac{\partial u}{\partial(x^2+y^2)} \begin{pmatrix} \frac{\partial(x^2+y^2)}{\partial x} & \frac{\partial(x^2+y^2)}{\partial y} \end{pmatrix} = \frac{\partial u}{\partial(x^2+y^2)} (2x \quad 2y)$$

$$\Rightarrow y \frac{\partial u}{\partial x} - x \frac{\partial u}{\partial y} = \left(\frac{\partial u}{\partial x} \quad \frac{\partial u}{\partial y} \right) \begin{pmatrix} y \\ -x \end{pmatrix} = \frac{\partial u}{\partial(x^2+y^2)} (2x \quad 2y) \begin{pmatrix} y \\ -x \end{pmatrix} = 0$$

2.

$$x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} = \left(\frac{\partial u}{\partial x} \quad \frac{\partial u}{\partial y} \right) \begin{pmatrix} x \\ -y \end{pmatrix} = \frac{\partial u}{\partial(xy)} (y \quad x) \begin{pmatrix} x \\ -y \end{pmatrix} = 0$$

$$3. x \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} = \left(\frac{\partial u}{\partial x} \quad \frac{\partial u}{\partial y} \right) \begin{pmatrix} x \\ y^2 \end{pmatrix} = \frac{\partial u}{\partial(\log x + \frac{1}{y})} \left(\frac{1}{x} \quad -\frac{1}{y^2} \right) \begin{pmatrix} x \\ y^2 \end{pmatrix} = 0$$

$$4. \psi'(y) \frac{\partial u}{\partial x} - \varphi'(x) \frac{\partial u}{\partial y} = \left(\frac{\partial u}{\partial x} \quad \frac{\partial u}{\partial y} \right) \begin{pmatrix} \psi'(y) \\ -\varphi'(x) \end{pmatrix} = \frac{\partial u}{\partial(\varphi(x) + \psi(y))} (\varphi'(x) \quad \psi'(y)) \begin{pmatrix} \psi'(y) \\ -\varphi'(x) \end{pmatrix} = 0$$

$$\Leftrightarrow \psi'(y) \frac{\partial u}{\partial x} = \varphi'(x) \frac{\partial u}{\partial y}$$

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5. (1) 設 $\varepsilon = x+y$, $\eta = xy$. $u = f(\varepsilon, \eta)$

$$\begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial \varepsilon} & \frac{\partial f}{\partial \eta} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ y & x \end{pmatrix} = \left(\frac{\partial f}{\partial \varepsilon} + y \frac{\partial f}{\partial \eta} \quad \frac{\partial f}{\partial \varepsilon} + x \frac{\partial f}{\partial \eta} \right)$$

$$(2) \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial (xy)} & \frac{\partial f}{\partial (xyz)} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ y & x & 0 \\ yz & zx & xy \end{pmatrix}$$

$$= \left(\frac{\partial f}{\partial x} + y \frac{\partial f}{\partial (xy)} + yz \frac{\partial f}{\partial (xyz)} \quad x \frac{\partial f}{\partial (xy)} + zx \frac{\partial f}{\partial (xyz)} \quad xy \frac{\partial f}{\partial (xyz)} \right)$$

$$(3) \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial (\frac{x}{y})} & \frac{\partial f}{\partial (\frac{y}{z})} \end{pmatrix} \begin{pmatrix} \frac{1}{y} & -\frac{x}{y^2} & 0 \\ 0 & \frac{1}{z} & -\frac{y}{z^2} \end{pmatrix}$$

$$= \left(\frac{1}{y} \frac{\partial f}{\partial (\frac{x}{y})} \quad -\frac{x}{y^2} \frac{\partial f}{\partial (\frac{x}{y})} + \frac{1}{z} \frac{\partial f}{\partial (\frac{y}{z})} \quad -\frac{y}{z^2} \frac{\partial f}{\partial (\frac{y}{z})} \right)$$

6.

因為 $0 = \frac{\partial u}{\partial x} - x = \frac{\partial}{\partial x} (u - \frac{1}{2}x^2)$, 所以 $\exists g: \mathbb{R} \rightarrow \mathbb{R}$, 使得 $g(y) = u - \frac{1}{2}x^2$

由題意知 $g(x^2) = 1 - \frac{1}{2}x^2$ (當 $y = x^2$ 時), 即此時有 $g(y) = 1 - \frac{1}{2}y$

$$\frac{\partial u}{\partial y} \Big|_{y=x^2} = \frac{dg}{dy} \Big|_{y=x^2} = -\frac{1}{2}$$

$$7. \begin{pmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \end{pmatrix} \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = (2xy \ y^2) \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}$$

$$= (2xy \cos \theta + y^2 \sin \theta \quad -2rxy \sin \theta + y^2 r \cos \theta) = (2r^2 \cos^2 \theta \sin \theta + r^2 \sin^4 \theta \quad -2r^3 \cos \theta \sin^2 \theta + r^3 \sin^2 \theta \cos \theta)$$

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8. 由微分定義知, 取 $\vec{h} = t(x, y, z)$, $\vec{R} = t(u, v, w)$ 記 $\vec{X} = (x, y, z)$, $\vec{U} = (u, v, w)$

$$f(\vec{X} + \vec{h}) - f(\vec{X}) = \left\langle \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right), \vec{h} \right\rangle + o(\|\vec{h}\|)$$

$$\Leftrightarrow f((1+t)\vec{X}) - f(\vec{X}) = t \left(x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} \right) + o(|t|)$$

不難發現 $\forall \alpha \in \mathbb{R}$, 有 $f(\alpha \vec{X}) = F(\alpha \vec{U})$

$$\text{因此 } F((1+t)\vec{U}) - F(\vec{U}) = \left\langle \left(\frac{\partial F}{\partial u}, \frac{\partial F}{\partial v}, \frac{\partial F}{\partial w} \right), \vec{R} \right\rangle + o(|t|)$$

$$\text{同時, } F((1+t)\vec{U}) - F(\vec{U}) = t \left(u \frac{\partial F}{\partial u} + v \frac{\partial F}{\partial v} + w \frac{\partial F}{\partial w} \right) + o(|t|)$$

$$\Rightarrow x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} = u \frac{\partial F}{\partial u} + v \frac{\partial F}{\partial v} + w \frac{\partial F}{\partial w}$$

9. (1) 設 $(s_0, t_0) = (2, 1)$, $(x_0, y_0) = g(s_0, t_0) = (3, 3)$

$$J(\vec{f} \circ \vec{g})(s_0, t_0) = \begin{pmatrix} \frac{\partial x}{\partial s} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial s} & \frac{\partial y}{\partial t} \\ \frac{\partial(x^2 y)}{\partial s} & \frac{\partial(x^2 y)}{\partial t} \end{pmatrix}_{(x,y)=(x_0,y_0)} \begin{pmatrix} \frac{\partial(s+t)}{\partial s} & \frac{\partial(s+t)}{\partial t} \\ \frac{\partial(s^2-t^2)}{\partial s} & \frac{\partial(s^2-t^2)}{\partial t} \end{pmatrix}_{(s,t)=(s_0,t_0)}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 2x_0 y_0 & x_0^2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2s_0 & -2t_0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2s_0 & -2t_0 \\ 2x_0(y_0 + x_0 s_0) & 2x_0(y_0 - x_0 t_0) \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 1 \\ 4 & -2 \\ 54 & 0 \end{pmatrix}$$

$$(2) J(\vec{f} \circ \vec{g}) = \begin{pmatrix} \frac{\partial \varphi}{\partial x}(x+y) & \frac{\partial \varphi}{\partial y}(x+y) \\ \frac{\partial \varphi}{\partial x}(x-y) & \frac{\partial \varphi}{\partial y}(x-y) \end{pmatrix}_{(x,y)=(e^t, e^{-t})} \begin{pmatrix} \frac{\partial e^t}{\partial s} & \frac{\partial e^t}{\partial t} \\ \frac{\partial e^{-t}}{\partial s} & \frac{\partial e^{-t}}{\partial t} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & e^t \frac{\partial \varphi}{\partial e^t}(e^t + e^{-t}) - e^{-t} \frac{\partial \varphi}{\partial e^{-t}}(e^t + e^{-t}) \\ 0 & e^t \frac{\partial \varphi}{\partial e^t}(e^t - e^{-t}) - e^{-t} \frac{\partial \varphi}{\partial e^{-t}}(e^t - e^{-t}) \end{pmatrix} = \begin{pmatrix} 0 & \frac{d\varphi}{dt}(e^t + e^{-t}) \\ 0 & \frac{d\varphi}{dt}(e^t - e^{-t}) \end{pmatrix}$$

$$\begin{aligned}
 (3) \quad J(\vec{f} \circ \vec{g}) &= \begin{pmatrix} 2x & 1 & 1 \\ 2 & 1 & 2z \\ 0 & 0 & 0 \end{pmatrix}_{(x,y,z) = \vec{g}(u,v,w)} \begin{pmatrix} v^2 w^2 & 2uvw^2 & 2uv^2 w \\ 0 & w^2 \cos v & 2w \sin v \\ 2u^2 e^v & u^2 e^v & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 2uv^2 w^2 & 1 & 1 \\ 2 & 1 & 2u^2 e^v \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v^2 w^2 & 2uvw^2 & 2uv^2 w \\ 0 & w^2 \cos v & 2w \sin v \\ 2u^2 e^v & u^2 e^v & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 2uv^4 w^4 + 2u^2 e^v & 2u^2 v^3 w^4 + w^2 \cos v + u^2 e^v & 4u^2 v^4 w^3 + 2w \sin v \\ 2v^2 w^2 + 4u^2 e^{2v} & 4uvw^2 + w^2 \cos v + 2u^4 e^{2v} & 4uv^2 w + 2w \sin v \\ 0 & 0 & 0 \end{pmatrix}
 \end{aligned}$$

10. 記 $E = (\vec{e}_1 \ \vec{e}_2 \ \vec{e}_3)$, 因為 E 是正交矩陣, 所以 $E^{-1} = E^T = \begin{pmatrix} \vec{e}_1^T \\ \vec{e}_2^T \\ \vec{e}_3^T \end{pmatrix}$

$$\frac{\partial f}{\partial \vec{e}_1} = \langle Jf(x,y,z), \vec{e}_1 \rangle = \left(\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \quad \frac{\partial f}{\partial z} \right) \vec{e}_1$$

同理可得 $\left(\frac{\partial f}{\partial \vec{e}_1} \quad \frac{\partial f}{\partial \vec{e}_2} \quad \frac{\partial f}{\partial \vec{e}_3} \right) = \left(\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \quad \frac{\partial f}{\partial z} \right) (\vec{e}_1 \ \vec{e}_2 \ \vec{e}_3) = \left(\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \quad \frac{\partial f}{\partial z} \right) E$

$$\begin{aligned}
 \Rightarrow \left(\frac{\partial f}{\partial \vec{e}_1} \right)^2 + \left(\frac{\partial f}{\partial \vec{e}_2} \right)^2 + \left(\frac{\partial f}{\partial \vec{e}_3} \right)^2 &= \left(\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \quad \frac{\partial f}{\partial z} \right) \begin{pmatrix} \frac{\partial f}{\partial \vec{e}_1} \\ \frac{\partial f}{\partial \vec{e}_2} \\ \frac{\partial f}{\partial \vec{e}_3} \end{pmatrix} = \left(\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \quad \frac{\partial f}{\partial z} \right) E E^T \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{pmatrix} \\
 &= \left(\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \quad \frac{\partial f}{\partial z} \right) \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{pmatrix} = \left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 + \left(\frac{\partial f}{\partial z} \right)^2
 \end{aligned}$$

練習題 14.6

$$1. (2) \quad 0 = y + x \frac{dy}{dx} - \frac{1}{y} \frac{dy}{dx} \Leftrightarrow \frac{dy}{dx} = \frac{y}{\frac{1}{y} - x} = \frac{y^2}{1 - xy}$$

$$(4) \quad x^y = y^x \Leftrightarrow x^{\frac{y}{x}} = y^{\frac{x}{y}} \Leftrightarrow$$

$$\Leftrightarrow y \log x = x \log y \Rightarrow \frac{dy}{dx} \log x + \frac{y}{x} = \log y + \frac{x}{y} \frac{dy}{dx}$$

$$\begin{aligned}
 \Leftrightarrow \left(\log x - \frac{x}{y} \right) \frac{dy}{dx} &= \log y - \frac{y}{x} \Leftrightarrow \frac{dy}{dx} = \frac{\log y - \frac{y}{x}}{\log x - \frac{x}{y}} = \frac{yx \log y - y^2}{xy \log x - x^2} \\
 &= \frac{y^2 (\log x - 1)}{x (y \log x - x)}
 \end{aligned}$$

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$$2. (2) \frac{x}{z} - \log z + \log y = 0$$

$$\Rightarrow \frac{x - x \frac{\partial z}{\partial x}}{z^2} - \frac{1}{z} \cdot \frac{\partial z}{\partial x} = 0 \Leftrightarrow z - x \frac{\partial z}{\partial x} - z \frac{\partial z}{\partial x} = 0$$

$$\Leftrightarrow \frac{\partial z}{\partial x} = \frac{z}{x+z}$$

$$\text{同理有 } -\frac{x}{z^2} \frac{\partial z}{\partial y} - \frac{1}{z} \frac{\partial z}{\partial y} + \frac{1}{y} = 0 \Leftrightarrow \frac{1}{y} = \left(\frac{x}{z^2} + \frac{1}{z} \right) \frac{\partial z}{\partial y}$$

$$\Leftrightarrow \frac{\partial z}{\partial y} = \frac{z^2}{y(x+z)}$$

$$(3) e^{x+y+z} (x+y+z) = 1$$

$$\Rightarrow e^y \left[e^{x+z} \cdot \left(1 + \frac{\partial z}{\partial x} \right) (x+y+z) + e^{x+z} \left(1 + \frac{\partial z}{\partial x} \right) \right] = 0$$

$$\Rightarrow e^{x+z} \left(1 + \frac{\partial z}{\partial x} \right) (x+y+z+1) = 0 \Leftrightarrow \frac{\partial z}{\partial x} = -1 \text{ 或 } x+y+z = -1 \text{ (捨)}$$

$$\text{同理可得 } \frac{\partial z}{\partial y} = -1. \text{ 由此可得 } z = -x - y + C$$

$$\Rightarrow x+y+z = C_0$$

$$\text{考慮函數 } f(t) = te^t - 1 \Rightarrow f'(t) = (1+t)e^t$$

$$\text{現在 } \begin{cases} Ce^C - 1 = 0 & \text{--- (i)} \\ f(C) = (1+C)e^C & \text{--- (ii)} \end{cases}$$

$$\text{由 (i) 知 } C > 0 \Rightarrow f'(C) > e^C > 0$$

$$\text{又知 } f(0) = -1 < 0, f(1) = e - 1 > 0$$

由零值定理和 f 在 $(-\infty, +\infty)$ 上的遞增性知 $\exists C_0 \in (0, +\infty)$

$$\text{使得 } C_0 e^{C_0} = 1$$

即 $x+y+z = C_0$, 因此原方程的解在 $\mathbb{R}^3$ 內的一個平面。

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$$(4) \quad z^2y - xz^3 - 1 = 0 \Rightarrow 2yz \frac{\partial z}{\partial x} - z^3 - 3xz^2 \frac{\partial z}{\partial x} = 0$$

$$\Rightarrow \frac{\partial z}{\partial x} = \frac{z^2}{2y - 3xz} \Rightarrow \left. \frac{\partial z}{\partial x} \right|_{(x,y,z)=(1,2,1)} = 1$$

$$\textcircled{1} \Rightarrow 2yz \frac{\partial z}{\partial y} + z^2 - 3xz^2 \frac{\partial z}{\partial y} = 0$$

$$\Rightarrow \frac{\partial z}{\partial y} = \frac{z}{3xz - 2y} \Rightarrow \left. \frac{\partial z}{\partial y} \right|_{(x,y,z)=(1,2,1)} = -1$$

$$3. \quad \frac{\partial x}{\partial y} = - \frac{\partial F / \partial y}{\partial F / \partial x}, \quad \frac{\partial y}{\partial z} = - \frac{\partial F / \partial z}{\partial F / \partial y}, \quad \frac{\partial z}{\partial x} = - \frac{\partial F / \partial x}{\partial F / \partial z}$$

$$\Rightarrow \frac{\partial x}{\partial y} \frac{\partial y}{\partial z} \frac{\partial z}{\partial x} = -1$$

$$4. \quad \text{記 } \alpha = x - y, \beta = y - z, \gamma = z - x$$

$$0 = \frac{\partial F}{\partial \alpha} + \frac{\partial F}{\partial \beta} \left(- \frac{\partial z}{\partial x} \right) + \frac{\partial F}{\partial \gamma} \left(\frac{\partial z}{\partial x} - 1 \right), \quad 0 = \frac{\partial F}{\partial \alpha} (0 - 1) + \frac{\partial F}{\partial \beta} \left(1 - \frac{\partial z}{\partial y} \right) + \frac{\partial F}{\partial \gamma} \left(\frac{\partial z}{\partial y} \right)$$

$$\Rightarrow \frac{\partial z}{\partial x} = \frac{\frac{\partial F}{\partial \alpha} - \frac{\partial F}{\partial \gamma}}{\frac{\partial F}{\partial \beta} - \frac{\partial F}{\partial \gamma}}, \quad \frac{\partial z}{\partial y} = \frac{\frac{\partial F}{\partial \beta} - \frac{\partial F}{\partial \alpha}}{\frac{\partial F}{\partial \beta} - \frac{\partial F}{\partial \gamma}}$$

$$5. \quad \text{記 } \alpha = x + y + z, \beta = x^2 + y^2 + z^2$$

$$0 = \frac{\partial F}{\partial \alpha} \left(1 + \frac{\partial z}{\partial x} \right) + \frac{\partial F}{\partial \beta} \left(2x + 2z \frac{\partial z}{\partial x} \right) \Rightarrow \frac{\partial z}{\partial x} = - \frac{\frac{\partial F}{\partial \alpha} + 2x \frac{\partial F}{\partial \beta}}{\frac{\partial F}{\partial \alpha} + 2z \frac{\partial F}{\partial \beta}}$$

$$\text{由 } x, y \text{ 的對稱性知, } \frac{\partial z}{\partial y} = - \frac{\frac{\partial F}{\partial \alpha} + 2y \frac{\partial F}{\partial \beta}}{\frac{\partial F}{\partial \alpha} + 2z \frac{\partial F}{\partial \beta}}$$

練習題 14.7

1. 記 $F_1 = x^2 + y^2 + z^2 - 1$, $\vec{F} = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix}$
 $F_2 = x + y + z$

$$\vec{J}\vec{F} = \begin{pmatrix} 2x & 2y & 2z \\ 1 & 1 & 1 \end{pmatrix}, \begin{pmatrix} \frac{dy}{dx} \\ \frac{dz}{dx} \end{pmatrix} = - \begin{pmatrix} 2y & 2z \\ 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 2x \\ 1 \end{pmatrix} = \frac{1}{2(z-y)} \begin{pmatrix} 1 & -1 \\ -2z & 2y \end{pmatrix} \begin{pmatrix} 2x \\ 1 \end{pmatrix}$$

$$= \frac{1}{2(z-y)} \begin{pmatrix} 1 & -2z \\ -1 & 2y \end{pmatrix} \begin{pmatrix} 2x \\ 1 \end{pmatrix} = \frac{1}{2(z-y)} \begin{pmatrix} 2(x-z) \\ 2(y-x) \end{pmatrix} = \frac{1}{z-y} \begin{pmatrix} x-z \\ y-x \end{pmatrix}$$

半徑為 1 的球面與平面 $x+y+z=0$ 的圓周

2. 記 $F_1 = x^2 + y^2 - z$, $\vec{F} = \begin{pmatrix} F_1 \\ F_2 \end{pmatrix}$
 $F_2 = x^2 + 2y^2 + 3z^2 - 20$

$$\vec{J}\vec{F} = \begin{pmatrix} 2x & 2y & -1 \\ 2x & 4y & 6z \end{pmatrix}, \begin{pmatrix} \frac{dy}{dx} \\ \frac{dz}{dx} \end{pmatrix} = - \begin{pmatrix} 2y & -1 \\ 4y & 6z \end{pmatrix}^{-1} \begin{pmatrix} 2x \\ 2x \end{pmatrix}$$

$$= - \frac{2x}{12yz+4y} \begin{pmatrix} 6z & -1 \\ -4y & 2y \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = - \frac{x}{6yz+2y} \begin{pmatrix} 6z-1 \\ -2y \end{pmatrix}$$

直角錐體與橢球面的交集

3. $y = t^2 + \frac{1}{t} = (t + \frac{1}{t})^2 - 2 = x^2 - 2 \Rightarrow \frac{dy}{dx} = 2x$

$$z = t^3 + \frac{1}{t} = (t + \frac{1}{t})^3 - 3t \cdot \frac{1}{t} (t + \frac{1}{t}) = x^3 - 3x \Rightarrow \frac{dz}{dx} = 3x^2 - 3$$

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4. (1) 記 $\vec{F} = \begin{pmatrix} xu - yv \\ yu + xv - 1 \end{pmatrix}$, $J\vec{F} = \begin{pmatrix} u & -v & x & -y \\ v & u & y & x \end{pmatrix}$

$$\begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = - \begin{pmatrix} u & -v \\ v & u \end{pmatrix}^{-1} \begin{pmatrix} x & -y \\ y & x \end{pmatrix} = - \frac{1}{u^2 + v^2} \begin{pmatrix} u & v \\ -v & u \end{pmatrix} \begin{pmatrix} x & -y \\ y & x \end{pmatrix}$$

$$= - \frac{1}{u^2 + v^2} \begin{pmatrix} xu + yv & xv - yu \\ yu - xv & xu + yv \end{pmatrix}$$

(2) 記 $\vec{F} = \begin{pmatrix} x+y-u-v \\ \frac{x}{y} - \frac{\sin u}{\sin v} \end{pmatrix}$, $J\vec{F} = \begin{pmatrix} 1 & 1 & -1 & -1 \\ \frac{1}{y} & -\frac{x}{y^2} & -\frac{\cos u}{\sin v} & \frac{\sin u}{\sin^2 v} \cos v \end{pmatrix}$

$$\begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = - \begin{pmatrix} 1 & 1 \\ \frac{1}{y} & -\frac{x}{y^2} \end{pmatrix}^{-1} \begin{pmatrix} -1 & -1 \\ -\frac{\cos u}{\sin v} & \frac{\sin u}{\sin^2 v} \cos v \end{pmatrix}$$

$$= \frac{1}{\frac{x}{y^2} + \frac{1}{y}} \begin{pmatrix} -\frac{x}{y^2} & -1 \\ -\frac{1}{y} & 1 \end{pmatrix} \begin{pmatrix} -1 & -1 \\ -\frac{\cos u}{\sin v} & \frac{\sin u}{\sin^2 v} \cos v \end{pmatrix}$$

$$= \frac{y^2}{x+y} \begin{pmatrix} \frac{x}{y^2} + \frac{\cos u}{\sin v} & \frac{x}{y^2} - \frac{\sin u}{\sin^2 v} \cos v \\ \frac{1}{y} - \frac{\cos u}{\sin v} & \frac{1}{y} + \frac{\sin u}{\sin^2 v} \cos v \end{pmatrix}$$

5. 記 $\vec{F} = \begin{pmatrix} u^2 - v \cos(xy) + w^2 \\ u^2 + v^2 - \sin(xy) + 2w^2 - 2 \\ uv - \sin x \cos y + w \end{pmatrix}$

$$J\vec{F} = \begin{pmatrix} yv \sin(xy) & xv \sin(xy) & 2u & -\cos(xy) & 2w \\ -y \cos(xy) & -x \cos(xy) & 2u & 2v & 4w \\ -\cos y \cos x & \sin x \sin y & v & u & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 2 & -1 & 0 \\ 0 & -\frac{\pi}{2} & 2 & 2 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{pmatrix} = - \begin{pmatrix} 2 & -1 & 0 \\ 2 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 0 & 0 \\ 0 & -\frac{\pi}{2} \\ 0 & 0 \end{pmatrix} = \frac{\pi}{2} \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix} = \frac{\pi}{2} \begin{pmatrix} 0 & 1 \\ 0 & 2 \\ 0 & -3 \end{pmatrix}$$

6. 記 $\vec{F} = \begin{pmatrix} f(x, y, z, t) - u \\ g(y, z, t) \\ h(z, t) \end{pmatrix}$, $J\vec{F} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} & \frac{\partial f}{\partial t} & -1 \\ 0 & \frac{\partial g}{\partial y} & \frac{\partial g}{\partial z} & \frac{\partial g}{\partial t} & 0 \\ 0 & 0 & \frac{\partial h}{\partial z} & \frac{\partial h}{\partial t} & 0 \end{pmatrix}$

$$\begin{pmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial t}{\partial x} & \frac{\partial t}{\partial y} \\ \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \end{pmatrix} = - \begin{pmatrix} \frac{\partial f}{\partial z} & \frac{\partial f}{\partial t} & -1 \\ \frac{\partial g}{\partial z} & \frac{\partial g}{\partial t} & 0 \\ \frac{\partial h}{\partial z} & \frac{\partial h}{\partial t} & 0 \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ 0 & \frac{\partial g}{\partial y} \\ 0 & 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \end{pmatrix} = - \left(\begin{vmatrix} \frac{\partial g}{\partial z} & \frac{\partial g}{\partial t} \\ \frac{\partial h}{\partial z} & \frac{\partial h}{\partial t} \end{vmatrix} - \begin{vmatrix} \frac{\partial f}{\partial z} & \frac{\partial f}{\partial t} \\ \frac{\partial h}{\partial z} & \frac{\partial h}{\partial t} \end{vmatrix} \begin{vmatrix} \frac{\partial f}{\partial z} & \frac{\partial f}{\partial t} \\ \frac{\partial g}{\partial z} & \frac{\partial g}{\partial t} \end{vmatrix} \right) \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ 0 & \frac{\partial g}{\partial y} \\ 0 & 0 \end{pmatrix}$$

$$= - \left(\frac{\partial f}{\partial x} \begin{vmatrix} \frac{\partial g}{\partial z} & \frac{\partial g}{\partial t} \\ \frac{\partial h}{\partial z} & \frac{\partial h}{\partial t} \end{vmatrix} - \frac{\partial f}{\partial y} \begin{vmatrix} \frac{\partial g}{\partial z} & \frac{\partial g}{\partial t} \\ \frac{\partial h}{\partial z} & \frac{\partial h}{\partial t} \end{vmatrix} - \frac{\partial g}{\partial y} \begin{vmatrix} \frac{\partial f}{\partial z} & \frac{\partial f}{\partial t} \\ \frac{\partial h}{\partial z} & \frac{\partial h}{\partial t} \end{vmatrix} \right)$$

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練習題 14.8

$$2. (1) J_{\vec{f}} = \begin{pmatrix} 2x & 0 \\ \frac{y}{x^2} & \frac{1}{x} \end{pmatrix}, J_{\vec{f}}^{-1} = \frac{1}{2} \begin{pmatrix} \frac{1}{x} & 0 \\ \frac{y}{x^2} & 2x \end{pmatrix} = \begin{pmatrix} \frac{1}{2x} & 0 \\ \frac{y}{2x^2} & x \end{pmatrix}$$

$$(2) J_{\vec{f}} = \begin{pmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{pmatrix}, J_{\vec{f}}^{-1} = e^x \begin{pmatrix} \cos(-y) & -\sin(-y) \\ \sin(-y) & \cos(-y) \end{pmatrix} = \begin{pmatrix} e^x \cos y & e^x \sin y \\ -e^x \sin y & e^x \cos y \end{pmatrix}$$

$$(3) J_{\vec{f}} = \begin{pmatrix} 1 & 1 \\ 2y^2 & 4xy \end{pmatrix}, J_{\vec{f}}^{-1} = \frac{1}{4xy - 2y^2} \begin{pmatrix} 4xy & -1 \\ -2y^2 & 1 \end{pmatrix}$$

練習題 14.9

$$1. (1) z = xy + \frac{x}{y}$$

$$\text{解: } \frac{\partial z}{\partial x} = y + \frac{1}{y}, \quad \frac{\partial z}{\partial y} = x - \frac{x}{y^2}$$

$$\frac{\partial^2 z}{\partial x^2} = 0, \quad \frac{\partial^2 z}{\partial y \partial x} = 1 - \frac{1}{y^2}, \quad \frac{\partial^2 z}{\partial x \partial y} = 1 - \frac{1}{y^2}, \quad \frac{\partial^2 z}{\partial y^2} = \frac{2x}{y^3}$$

$$(3) z = \log(x^2 + y^2)$$

$$\text{解: } \frac{\partial z}{\partial x} = \frac{2x}{x^2 + y^2}, \quad \frac{\partial z}{\partial y} = \frac{2y}{x^2 + y^2}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{2(x^2 + y^2) - 4x^2}{(x^2 + y^2)^2} = \frac{2(y^2 - x^2)}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{2(x^2 - y^2)}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y} = \frac{4xy}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 u}{\partial z \partial y} = \frac{\partial^2 u}{\partial y \partial z} = x (\cos(xy z) - xy z \sin(xy z))$$

$$\frac{\partial^2 u}{\partial z \partial x} = \frac{\partial^2 u}{\partial x \partial z} = y (\cos(xy z) - xy z \sin(xy z))$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x} = z \cos(xy z) - xy z^2 \sin(xy z)$$

$$(5) u = \sin(xy z)$$

$$\text{解: } \frac{\partial u}{\partial x} = y z \cos(xy z), \quad \frac{\partial u}{\partial y} = z x \cos(xy z), \quad \frac{\partial u}{\partial z} = xy \cos(xy z)$$

$$\frac{\partial^2 u}{\partial x^2} = -y^2 z^2 \sin(xy z), \quad \frac{\partial^2 u}{\partial y^2} = -z^2 x^2 \sin(xy z), \quad \frac{\partial^2 u}{\partial z^2} = -x^2 y^2 \sin(xy z)$$

(8) $u = x^{yz}$

解. $u = x^{yz} = e^{yz \ln x}$, $\frac{\partial u}{\partial x} = yz x^{yz-1}$, $\frac{\partial u}{\partial y} = x^{yz} \cdot z \ln x$, $\frac{\partial u}{\partial z} = x^{yz} \cdot y \ln x$

$$\frac{\partial^2 u}{\partial x^2} = yz(yz-1)x^{yz-2}, \quad \frac{\partial^2 u}{\partial y^2} = z^2(\ln x)^2 x^{yz}, \quad \frac{\partial^2 u}{\partial z^2} = y^2(\ln x)^2 x^{yz}$$

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y} = z(x^{yz-1} \cdot yz \ln x + x^{yz-1}) = z x^{yz-1} (yz \ln x + 1)$$

$$\frac{\partial^2 u}{\partial z \partial x} = \frac{\partial^2 u}{\partial x \partial z} = y x^{yz-1} (yz \ln x + 1)$$

$$\frac{\partial^2 u}{\partial z \partial y} = \frac{\partial^2 u}{\partial y \partial z} = \ln x (x^{yz} + y \cdot x^{yz} \cdot \ln x \cdot z) = x^{yz} \ln x (yz \ln x + 1)$$

(9) $u = \log(x_1 + x_2 + \dots + x_n)$

解. $\frac{\partial u}{\partial x_i} = \frac{1}{x_1 + x_2 + \dots + x_n}$, $\frac{\partial^2 u}{\partial x_i \partial x_j} = -\frac{1}{(x_1 + x_2 + \dots + x_n)^2}$

$$i, j \in \{1, 2, \dots, n\}$$

(10) $u = \arcsin(x_1^2 + x_2^2 + \dots + x_n^2)$

解. $\frac{\partial u}{\partial x_i} = \frac{2x_i}{\sqrt{1 - (x_1^2 + x_2^2 + \dots + x_n^2)}} = \frac{2x_i}{\sqrt{1 - \sin^2 u}} = \frac{2x_i}{\cos u}$

$$\frac{\partial^2 u}{\partial x_i \partial x_j} = \frac{2}{\cos u} \cdot (-\sin u) \cdot \frac{\partial u}{\partial x_j} = -\frac{2x_i}{\cos^2 u} \cdot (-\sin u) \cdot \frac{\partial u}{\partial x_j}$$

$$= \frac{\sin u}{1 - \sin^2 u} \cdot 2x_i \cdot \frac{2x_j}{\cos u} = \frac{4 \sum_{k=1}^n x_k^2}{(1 - (\sum_{k=1}^n x_k^2)^2)^{\frac{3}{2}}} x_i x_j$$

2. $\frac{\partial u}{\partial x} = \frac{-\frac{y}{x^2}}{1 + (\frac{y}{x})^2} = -\frac{y}{x^2 + y^2}$, $\frac{\partial u}{\partial y} = \frac{\frac{1}{x}}{1 + (\frac{y}{x})^2} = \frac{x}{x^2 + y^2}$

$\frac{\partial^2 u}{\partial x^2} = \frac{2xy}{(x^2 + y^2)^2}$, $\frac{\partial^2 u}{\partial y^2} = -\frac{2xy}{(x^2 + y^2)^2}$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

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3. $u = e^{a\theta} \cos(a \log r)$

解 $\frac{\partial^2 u}{\partial r^2} = e^{a\theta} (-\sin(a \log r)) \cdot \frac{a}{r} = -\frac{a \cdot e^{a\theta}}{r} \sin(a \log r)$

$\frac{\partial u}{\partial \theta} = a e^{a\theta} \cos(a \log r), \quad \frac{\partial^2 u}{\partial \theta^2} = a^2 e^{a\theta} \cos(a \log r) = a^2 u$

$\frac{\partial^2 u}{\partial r^2} = -a \cdot e^{a\theta} \cdot \cos(a \log r) \cdot \frac{a}{r} \cdot r - \sin(a \log r)$
 $= -a \cdot u \cdot a - \frac{e^{a\theta} \sin(a \log r)}{r^2} = -\frac{1}{r^2} a^2 u + \frac{1}{r} \left(-\frac{\partial u}{\partial r} \right)$

$\Rightarrow \frac{\partial^2 u}{\partial r^2} + \frac{1}{r^2} a^2 u + \frac{1}{r} \frac{\partial u}{\partial r} = 0 \Leftrightarrow \frac{\partial^2 u}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{1}{r} \frac{\partial u}{\partial r} = 0$

5. $\frac{\partial u}{\partial t} = \frac{1}{p} \left(\frac{\partial \psi}{\partial \varepsilon} \frac{\partial \varepsilon}{\partial t} + \frac{\partial \psi}{\partial \eta} \frac{\partial \eta}{\partial t} \right), \quad \varepsilon = p - at, \quad \eta = p + at$

$= \frac{a}{p} \left(\frac{\partial \psi}{\partial \eta} - \frac{\partial \psi}{\partial \varepsilon} \right), \quad \frac{\partial^2 u}{\partial t^2} = \frac{a}{p} \left(\frac{\partial^2 \psi}{\partial \eta^2} \cdot a - \frac{\partial^2 \psi}{\partial \varepsilon^2} (-a) \right) = \frac{a^2}{p} \left(\frac{\partial^2 \psi}{\partial \varepsilon^2} + \frac{\partial^2 \psi}{\partial \eta^2} \right)$

$\frac{\partial u}{\partial x} = \frac{1}{p^2} \left(\frac{\partial \psi}{\partial \varepsilon} \frac{\partial \varepsilon}{\partial x} + \frac{\partial \psi}{\partial \eta} \frac{\partial \eta}{\partial x} \right) p - (\psi(\varepsilon) + \psi(\eta)) \frac{\partial p}{\partial x}$

$= \frac{1}{p^2} \left(\left(\frac{\partial \psi}{\partial \varepsilon} + \frac{\partial \psi}{\partial \eta} \right) p - (\psi(\varepsilon) + \psi(\eta)) \right) \frac{\partial p}{\partial x} = \frac{x}{p^3} \left[\left(\frac{\partial \psi}{\partial \varepsilon} + \frac{\partial \psi}{\partial \eta} \right) p - (\psi(\varepsilon) + \psi(\eta)) \right]$

$\frac{\partial^2 u}{\partial x^2} = \frac{p^3 - 3x^2}{p^6} \cdot \frac{x}{p} \left[\left(\frac{\partial \psi}{\partial \varepsilon} + \frac{\partial \psi}{\partial \eta} \right) p - (\psi(\varepsilon) + \psi(\eta)) \right] + \frac{x}{p^3} \left[\left(\frac{\partial^2 \psi}{\partial \varepsilon^2} + \frac{\partial^2 \psi}{\partial \eta^2} \right) \frac{\partial p}{\partial x} \cdot p + \left(\frac{\partial \psi}{\partial \varepsilon} \cdot \frac{\partial \varepsilon}{\partial x} + \frac{\partial \psi}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} \right) \frac{\partial p}{\partial x} \right]$

$= \frac{p^3 - 3x^2}{p^5} \left[\left(\frac{\partial \psi}{\partial \varepsilon} + \frac{\partial \psi}{\partial \eta} \right) p - (\psi(\varepsilon) + \psi(\eta)) \right] + \frac{x^2}{p^3} \left[\left(\frac{\partial^2 \psi}{\partial \varepsilon^2} + \frac{\partial^2 \psi}{\partial \eta^2} \right) \right]$

$\Delta u = \frac{3p^3 - 3(x^2 + y^2 + z^2)}{p^5} \left[\left(\frac{\partial \psi}{\partial \varepsilon} + \frac{\partial \psi}{\partial \eta} \right) p - u p \right] + \frac{x^2 + y^2 + z^2}{p^3} \left[\frac{\partial^2 \psi}{\partial \varepsilon^2} + \frac{\partial^2 \psi}{\partial \eta^2} \right] = \frac{1}{p} \left(\frac{\partial^2 \psi}{\partial \varepsilon^2} + \frac{\partial^2 \psi}{\partial \eta^2} \right)$

$\Rightarrow \frac{\partial^2 u}{\partial t^2} = a^2 \cdot \frac{1}{p} \left(\frac{\partial^2 \psi}{\partial \varepsilon^2} + \frac{\partial^2 \psi}{\partial \eta^2} \right) = a^2 \Delta u$